Song-Ze Yu

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Research Interests

My research spans two parallel directions: grounded **music understanding** for audio-language models, and **controllable** music and audio-effect generation. My long-term goal is to advance audio-language models toward **real music intelligence**: the ability to hear, reason about, and interact with music.

Education

University of California, Berkeley Aug. 2025 - May 2026
Visiting Student in Computer Science *Advised by Prof. Trevor Darrell and Carmine-Emanuele Cella*
- GPA: 3.9 / 4.0

National Tsing Hua University Sep. 2023 - Jun. 2026
B.S. in Computer Science *Advised by Prof. Liu Yi-Wen*
- Overall GPA: 3.91 / 4.30

National Cheng Kung University Sep. 2022 - Jun. 2023
B.S. in Computer Science and Information Engineering
- Overall GPA: 4.16 / 4.30; highest department rank: **5 / 113**

Publications

(* indicates equal contribution; † indicates corresponding author)

[1] **Song-Ze Yu**†, Milan Liessens Dujardin, Yuxuan Cai, Wantong Zhang, Brian Cruz, Jeremy Wagner, Carmine-Emanuele Cella†, "*InstructFX2FX: A Multi-Turn Text-to-Effect System for Sequential Audio Effect Refinement*," DAFx 2026.

[2] Milan Liessens Dujardin*, **Song-Ze Yu***, Craver Corbyn Thomas-Smith, David M. Chan, Karina Nguyen, "*PitchBench: Measuring Pitch Hearing in Audio-Language Models*," NeurIPS 2026 D&B, under review.

[3] **Song-Ze Yu**, Joseph Suh, Serina Chang, David M. Chan, "*Anamnesis: An Open-Source Platform for Large-Scale Backstory-Conditioned Survey Simulation*," EMNLP 2026 Demo, under review.

[4] Milan Liessens Dujardin*, **Song-Ze Yu***, "*MuNo-LM: Unifying Score and Performance for Audio-Language Modeling*," TISMIR 2026, in preparation.

Research Experience

Music and AI Lab, National Taiwan University Jun. 2026 - Present
Research Assistant *Advised by Prof. Yi-Hsuan Yang*
- Improving **controllability in low-cost music generation** by aligning representations with music understanding.
- Post-training audio-language models to extend **PitchBench** from evaluating to teaching models musical perception.

Berkeley AI Research (BAIR) Lab, UC Berkeley Feb. 2026 - Present
Undergraduate Researcher *Advised by David M. Chan and Prof. Trevor Darrell*
- Co-developed **PitchBench**, a 28-experiment hierarchical benchmark showing that frontier audio-language models remain unreliable at both absolute and relative pitch perception, from isolated tones to melodic context.
- Built **Anamnesis**, an open-source survey-simulation platform that scales the **Anthology** method to 30K+ narrative backstories; reproduced Pew Research and New Yorker studies and better matched real-world opinion distributions than persona-prompting baselines.

Center for New Music and Audio Technologies Lab, UC Berkeley Sep. 2025 - Jun. 2026
Undergraduate Research Lead *Advised by Prof. Carmine-Emanuele Cella*
- Formulated **sequential FX refinement** as a stateful multi-turn problem and built **InstructFX2FX**, pairing LLM planning with CLAP-guided optimization to update effect chains without losing prior instructions.
- Proposed **MuNo-LM (Music Notation for Language Models)**, a compact representation of performance-level MIDI and score-level intent that enables zero-shot generation of fine-grained, note-level, and time-grounded captions and QA pairs for audio-language models.

AHG Music Lab, National Tsing Hua University Jul. 2024 - May 2025
Undergraduate Researcher *Advised by Prof. Yi-Wen Liu*
- Developed the **VTR model**, a single-shot audio-to-EQ model trained on a ReaScript-generated piano dataset for tone replication (MSE: 0.0216); integrated it into a VST plugin at Positive Grid, then moved beyond single-shot control toward iterative effect refinement with **InstructFX2FX** at CNMAT.

Selected Industry Experience

BerkeleyTime Sep. 2025 - May 2026
Machine Learning Pod Lead *Spring 2026*

- Interviewed and formed a **10-member ML Pod**, matching researchers and engineers by background into pairs across three workstreams: semantic-search maintenance, recommendation systems, and enrollment-probability prediction.
- Set weekly milestones, tracked progress, and reviewed code while coordinating researcher-engineer pairs with designers to bridge model selection, interface design, and production deployment.
- Led development of a **new Explore page** integrating the course-recommendation system into BerkeleyTime's user experience (presentation).

Machine Learning & Full-Stack Engineer, Catalog Pod *Fall 2025*

- Built BGE-based semantic search as a FastAPI microservice backed by FAISS and deployed it behind a toggle with fuzzy search on the Catalog page for **122K+ monthly users**.
- Reduced GraphQL first-load latency from **25 seconds to 1 second** through pagination and Redis caching.
- Shipped **20+ pull requests** and resolved urgent on-call bugs involving cross-listed courses and ghost classes.

Positive Grid Jun. 2025 - Aug. 2025
Machine Learning Intern *Taipei, Taiwan*

- Served as Scrum Master for **Amp-AI-SaaS** and designed the **Multi-Agent Orchestration Harness** for software engineering across Positive Grid's audio products.
- Built **two JUCE-based VST plugins**, including one that integrated the **VTR model** through a PyTorch inference pipeline.
- Explored source separation and zero-shot virtual-amplifier modeling for music-industry applications.

Awards & Leadership

Claude × a16z × Berkeley M.E.T. Makeathon — Best Design Oct. 2025

- Built Haven in five hours, a mobile app that personalizes home-design recommendations from room scans.

Meichu Hackathon — 2nd Place, Google Group Oct. 2024

- Built Travelity, an AI travel planner using interests, personality, and real-time map and search data.

Taipei Hackathon — 3rd Place Sep. 2024

- Built ParkFlow, a WebView microservice for real-time parking information, navigation, and notifications; selected for integration into Taipei's official city app (**3M+ downloads**) and awarded NT$100,000.
- Produced a hackathon vlog with **1K+ views** and was invited by the Taipei City Government the following year to share the winning experience with Taipei residents.

SITCON Hackathon — Group Award Winner Jul. 2024

- Built ccDiary, an AI-assisted mental-health journal with emotion analysis and anonymous social features.

National Cheng Kung University — Campus Speaker Aug. 2023

- Delivered an invited talk for incoming students on writing music as a diary, then turned audience-submitted phrases and sentences on Slido into improvised songs performed live.

Selected Music Experience

Pianist, Composer & Producer Music CV | YouTube (7K subscribers)

- Performed **24 solo piano recitals** across Greater China and appeared on **10+ Chinese-language TV variety shows and New Year galas (春晚)** between 2009 and 2016.
- Shared the stage with **Jay Chou (周杰倫) three times** and appeared as a special guest at a concert by **Yan Jue (嚴爵)** at Taipei Arena, performing eight-hands piano with Bai-An (白安) and Cosmos People (宇宙人).
- Composed **20+ original songs with 20K+ cumulative views**; performs on piano/keys, lead vocal, cajon, and electric guitar, and arranges, mixes, and masters original work.